ZEALCORPS

Energy Management System

Real-time monitoring, intelligent control, and proof of savings

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Visibility

Energy use & equipment health live Dashboards, Customizable Energy Reports & Audit Reports

Control

Adjust Setpoints Chiller, AHU , FCU, HVLS & Lighting centrally Trends, Events & Alarms Cloud Connectivity

Governance

Zone based control Times Schedules Security logs & role-based access AI powered

Why
Zealcorps EMS

Most buildings lose money in two places:

- Running equipment when it is not needed
- Not spotting issues early.
- In addition, Zealcorps applies a unique AI-driven optimization strategy that converts the BMS into a mathematical model and produces advisory recommendations for clients to enhance efficiency, reliability, and overall energy performance.

Visibility

Lower bills through better operating hours and fewer unnecessary run-times

More comfortable spaces (temperature and humidity are visible and trackable)

Faster troubleshooting: live status and logs reduce guesswork

Audit-ready exports (CSV/PDF) for management, reporting, and compliance

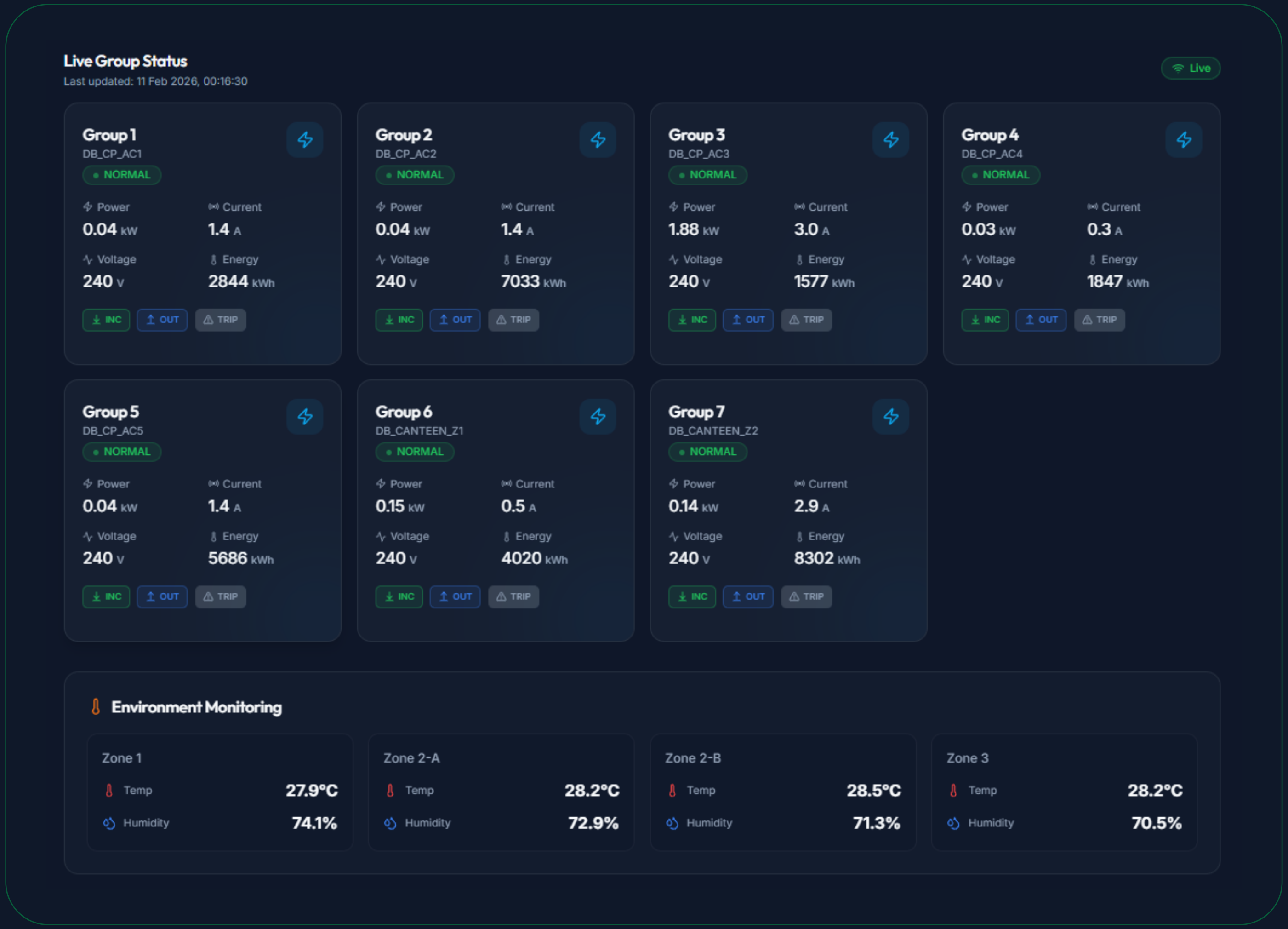
Dashboard: real-time monitoring and AI analysis

- Initial Automation
- Occupancy Sensors Added
- Zone Control Implementation
- Predictive AI Models
- Weather-Adaptive Systems
- Cloud AI Integration

Live Group Status: Operational impact

Measure & verify

- Higher kWh and peak demand charges
- Unplanned alarms and manual intervention
- Comfort / SLA risk when load changes quickly
- Hard to prove results to management



Live Group Status: Operational impact

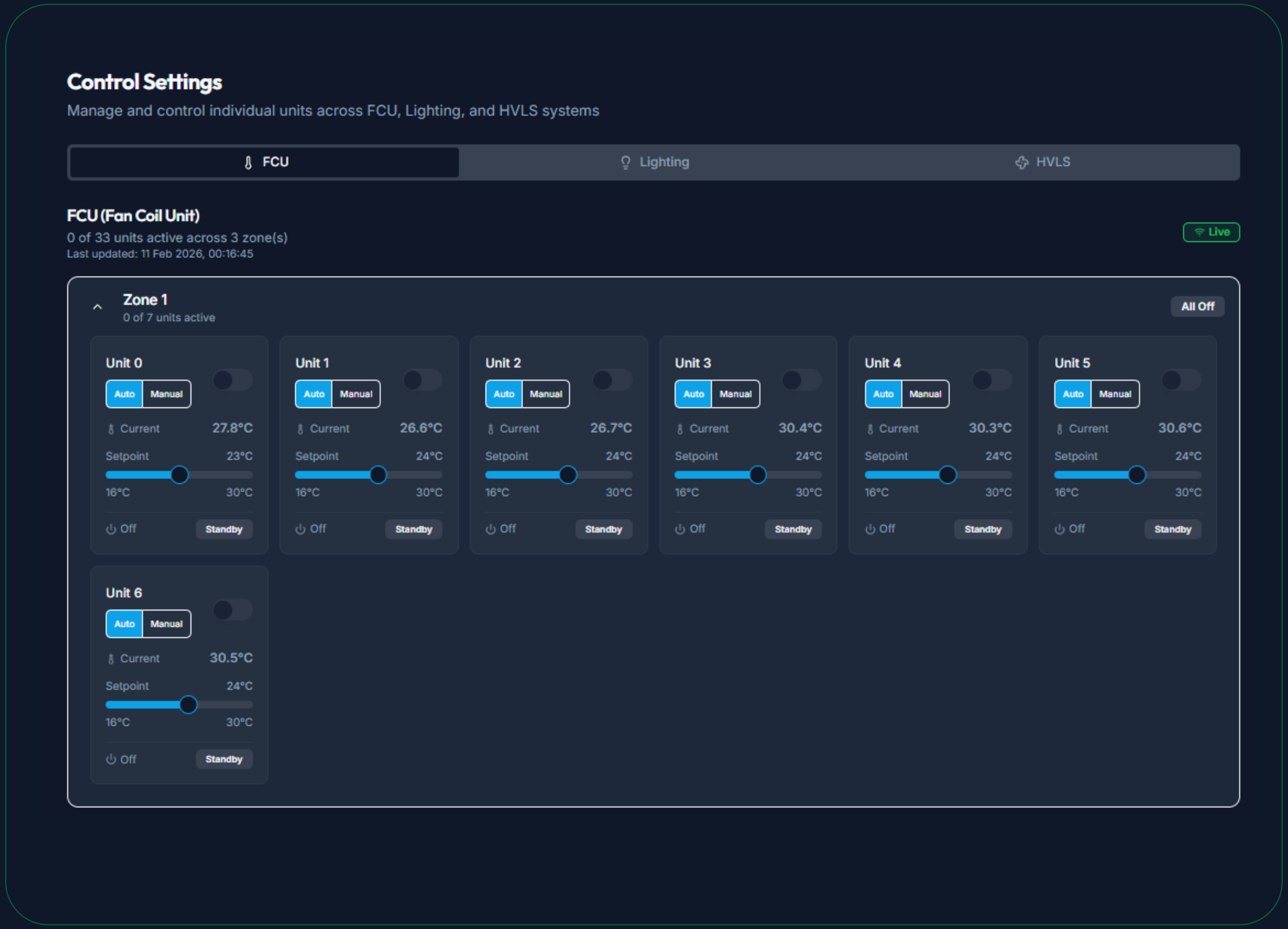
Dynamic setpoints

Imagine driving a car with cruise control:

Static: You always drive at 100 km/h no matter traffic, hills, or weather.

Dynamic: Your speed automatically adjusts based on traffic, road slope, and fuel efficiency.

Dynamic setpoints do the same thing for energy systems.



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Schedules: Smart Scheduling Groups

Schedule optimization

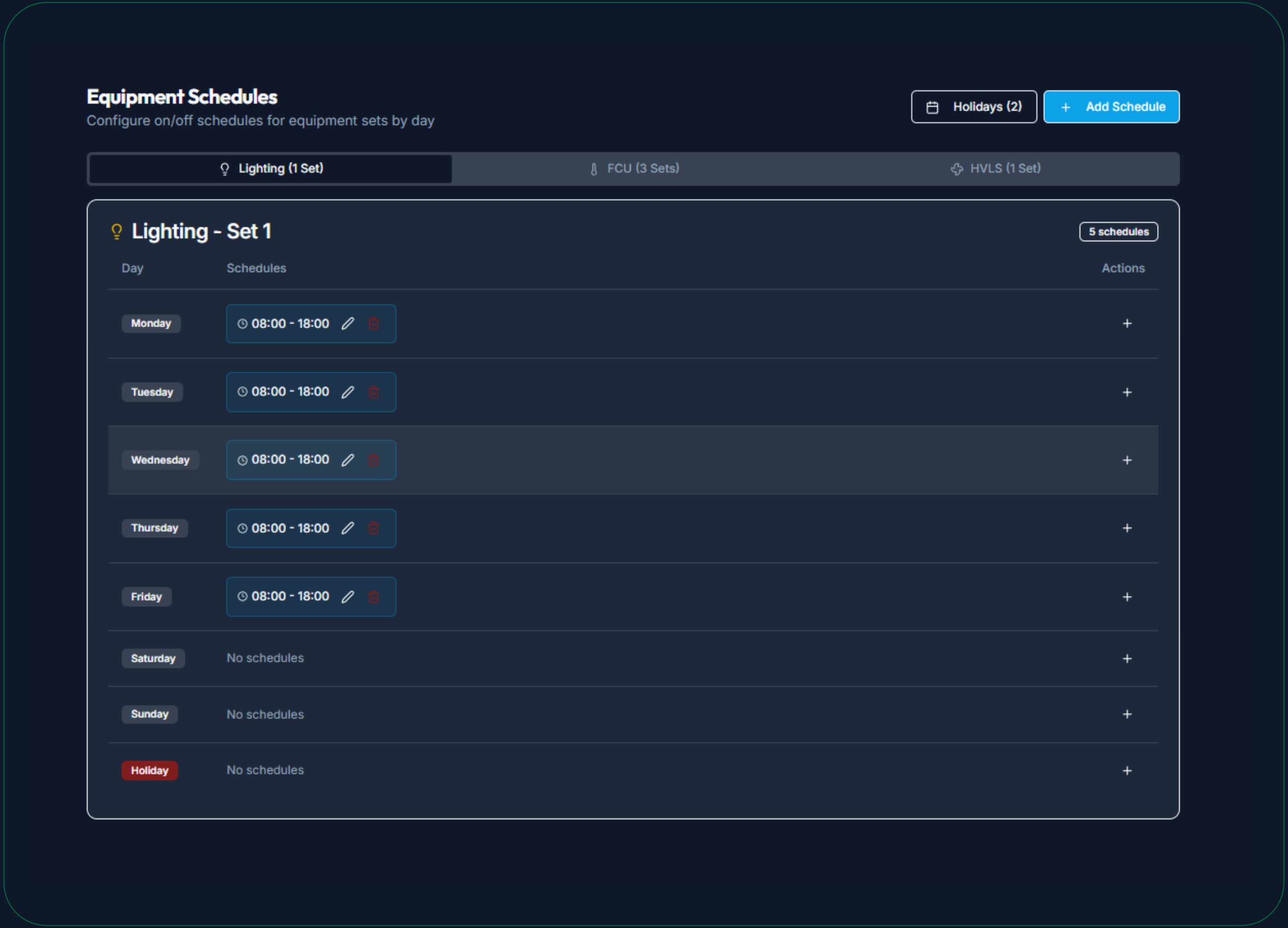
If demand is:

50% demand → Run Group A + partial Group B

75% demand → Run Group A + Group B

100% demand → Run all groups

Demand controls schedules. Schedules activate groups. Groups optimize efficiency.



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Reports: energy consumption with exportable Reports

Load forecasting

Predictive AI Models

Machine learning models started predicting thermal loads to proactively optimize energy consumption, improving efficiency.

Date	Hour	Energy (kWh)	Avg Power (kW)	Peak Power (kW)	Avg Voltage (V)	Peak Voltage (V)	Avg Current (A)
04/02/2026	06:00 - 07:00	0.00	64837719.85	85300027.00	3.0	3.0	8252.85
04/02/2026	07:00 - 08:00	0.00	44088822.49	99940077.00	3.0	3.0	19160.65
04/02/2026	08:00 - 09:00	0.00	55762746.76	98900068.00	3.0	3.0	19665.58
04/02/2026	09:00 - 10:00	0.00	51365509.38	97900096.00	3.0	3.0	23709.72
04/02/2026	10:00 - 11:00	0.00	53627678.36	99940090.00	3.0	3.0	24306.44
04/02/2026	11:00 - 12:00	0.00	54812427.33	99740083.00	3.0	3.0	23795.44

Zone Control reports: Weather-Adaptive Systems

Zone Control

zone-based controls, managing lighting and HVAC independently per to optimize energy use.

Adjusting setpoints dynamically based on real-time weather data for better temperature and fan control.

AI - Optimisation : ESG = Financial Advantage

Operations logs

Immediate Cost Reduction

Running the system at its most efficient operating state — while maintaining or improving comfort and reliability.

Predictable Financial Planning

“Based on conservative financial modeling using historical utility data, the system is structured to guarantee full payback within 24 months.”

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Summary of Energy Savings

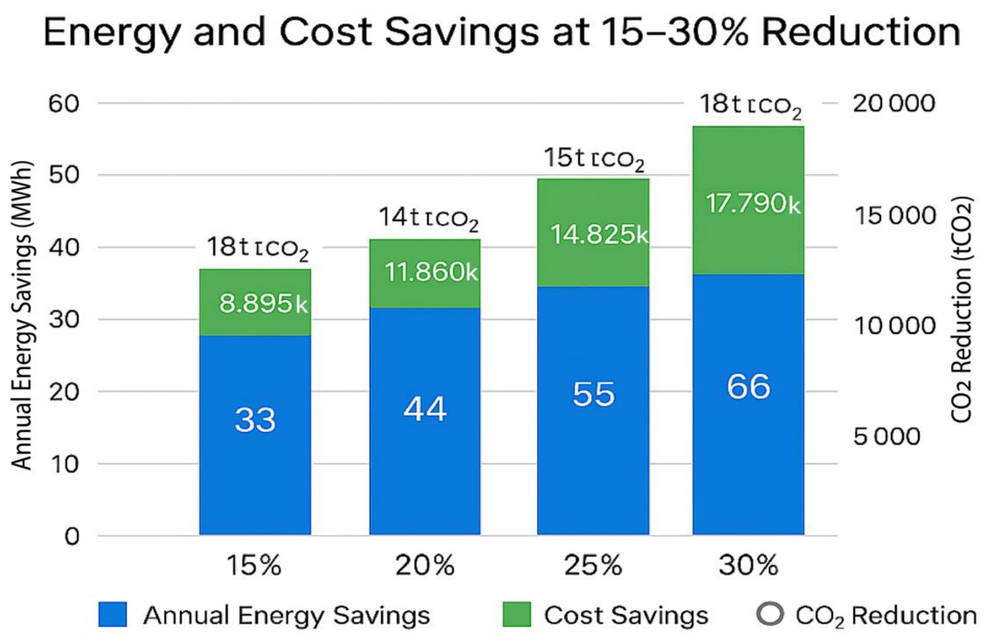
(Projected)

1. AI-driven time scheduling prevents unnecessary after-hours and weekend energy use, saving up to 10 %.

2. Zone-based controls tailor energy use per room type and occupancy, optimizing consumption by up to 8 %.

3. Occupancy sensors automate lighting and HVAC, eliminating waste in unoccupied spaces and saving up to 8 %.

4. Dynamic weather-adaptive adjustments ensure HVAC systems respond to real-time conditions, improving efficiency by up to 4 %



Saving %	Energy Saved (MWh)	Cost Saved (SGD)	CO_2 Reduction
15 %	33	8.895k	17.7
20%	44	11.860k	23.6
25%	55	14 825k	29.5
30%	66	17 790k	35.4

Average electricity tariff = 0.27 SGD/kWh

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Operations logs: Audit Trail = Accountability + Faster Investigations

AI Optimization Impact

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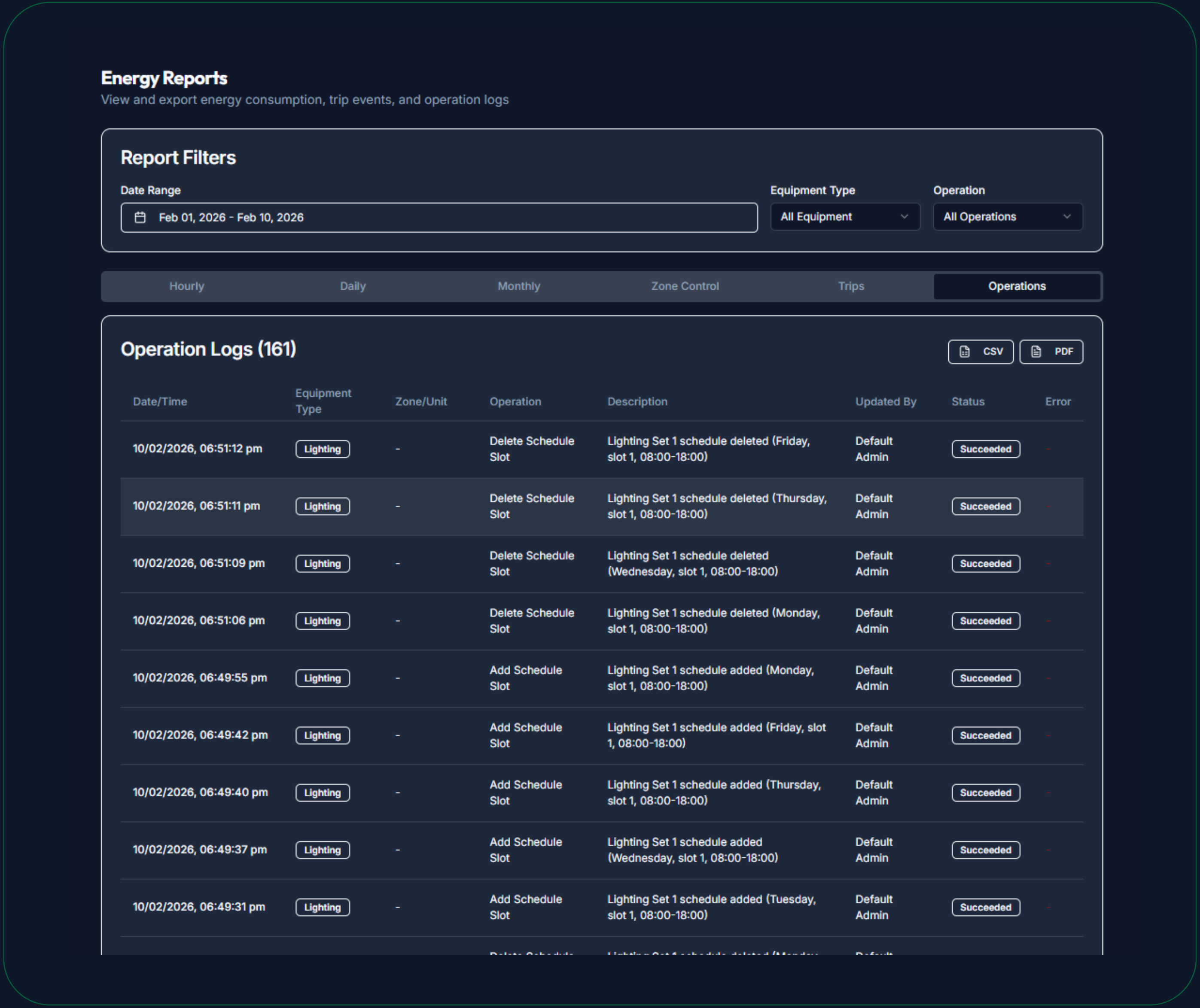

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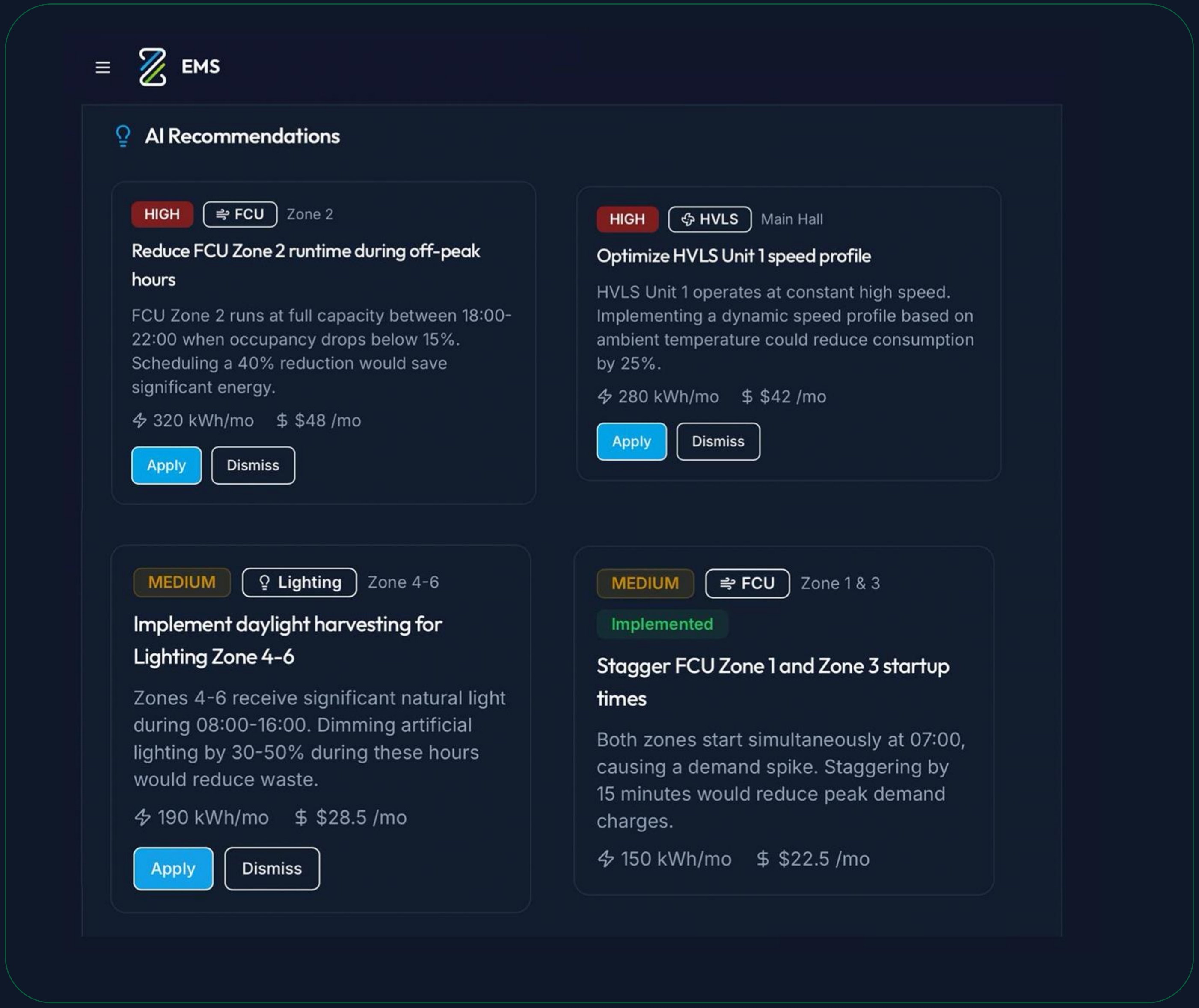
Intelligent Energy Recommendations

AI : Analyze + Recommend + Optimize

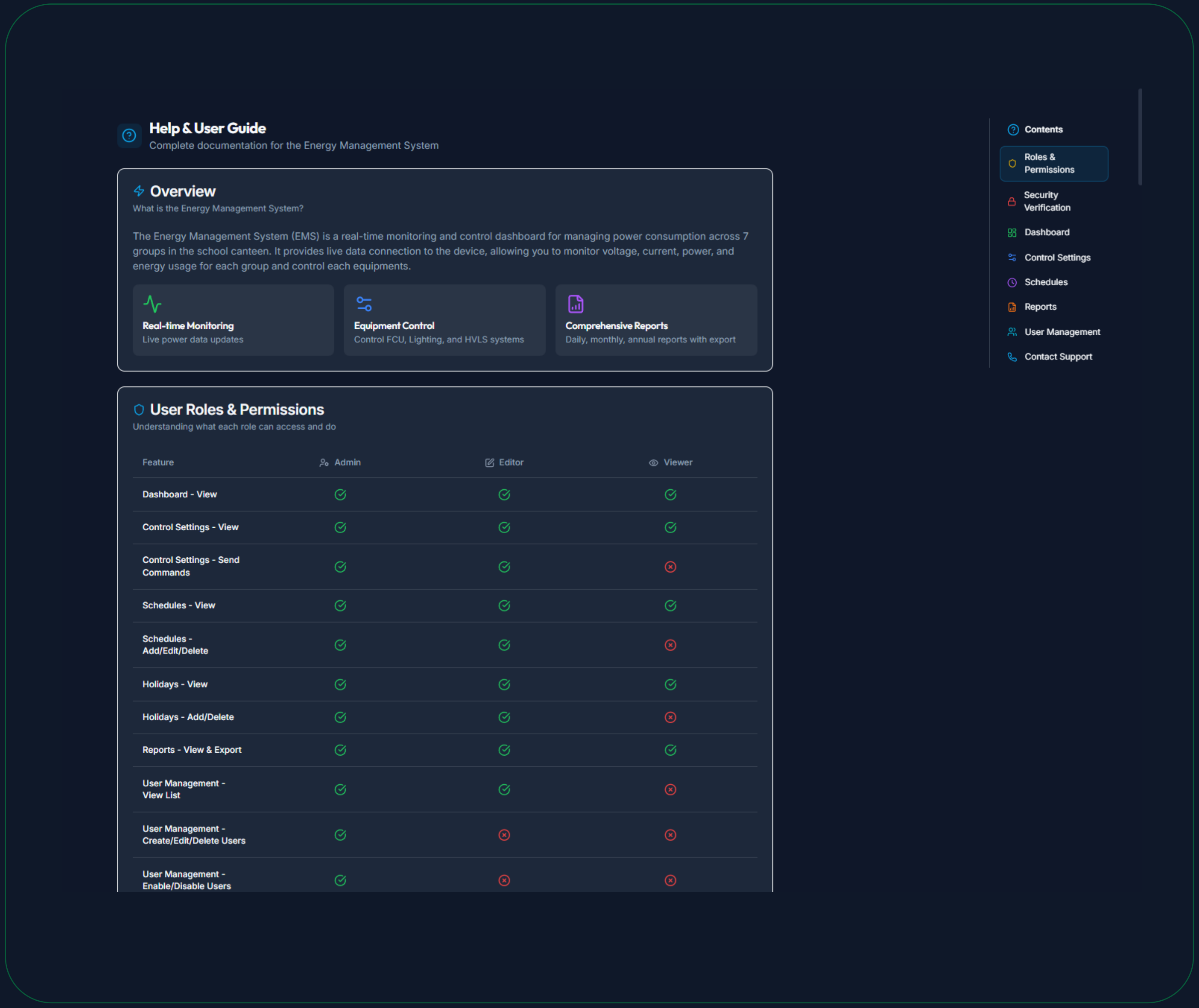
- AI evaluates equipment behavior and consumption trends across building systems.
- Smart alerts identify opportunities to reduce energy waste.
- Context-aware recommendations improve HVAC, lighting, and mechanical efficiency.
- Automated optimization suggestions help reduce operational costs.
- Clear visibility of potential venergy and cost savings for each recommendation.
- Continuous AI learning improves prediction accuracy over time.



Roles & permissions Security : Access control + audit + OT-safe design

AI Optimization Impact

- Cloud service hardened to ISO 27001 / SOC 2 / IM8
- TLS 1.3 encryption for MQTT/API
- AES-256 encryption at rest
- MFA + RBAC + SSO for all user and API access
- Key Vault Securely stores encryption keys, credentials, and API secrets used by EMS and AI.
- Azure Front Door Acts as global secure entry point with DDoS protection and load balancing.



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Connect with us and start unlocking greater savings with ZEMS.AI

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